Prediction of Defaulters using Machine Learning on Azure ML

Abhishek Shivanna
Center for Distributed and Mobile Computing, EECS
University of Cincinnati
Cincinnati, USA
shivanak@mail.uc.edu

Dharma P Agrawal
Center for Distributed and Mobile Computing, EECS
University of Cincinnati
Cincinnati, USA
dpa@cs.uc.edu

***Abstract*— Banks and lending institutions take risk in issuing new credit cards and loans to customers. Lending institutions at large need to have their own credit risk assessment system in accordance with Basel II guidelines. Many lending institutions lose a large amount of money as they do not have an accurate model to predict defaulters. The goal of credit risk management system is to accurately predict borrowers' ability to repay loans or make credit card payments in a timely manner. Researchers have taken multitude of approaches to solve this problem, and it continues to be an active area of research. Data mining and machine learning are emerging tools that are widely used by lending institutions to predict defaulters. These tools can effectively mine large dataset which is not feasible by traditional methods. In this work, we have used different algorithms including Deep Support Vector Machine (DSVM), Boosted Decision Tree (BDT), Averaged Perceptron (AP) and Bayes Point Machine (BPM) to build various models, in an attempt to better predict defaulters. Dataset, comprising of 25 attributes and 30k instances, was obtained from the repository of machine learning, University of California, Irvine (UCI). Our results show that, of all the four models used, DSVM can best predict defaulters. We believe that these models can be used to better predict defaulters by credit risk management system in banking and lending institutions.**

Keywords— Machine Learning; Pattern Recognition; Defaulter Prediction; Big Data; Azure ML; Data Mining; Perceptron; Decision Tree; SVM; Bayes Point Machine.

I. Introduction

The process of discovering useful information automatically from large repository is called data mining [1]. This technique employs various tools to explore available data, discover patterns, relationships and/or anomalies present in the data that were previously unknown. It gives us lot of insight into the data in hand [2]. The patterns we gather from data mining in-turn serve as inputs for various ML algorithms. There is widespread application of data mining in financial sector. In this research work, we have compared the performances of various ML classification algorithms to predict credit card defaulters. We have used the dataset from UCI's machine learning repository.

Basel II accord states that credit risks faced by banks and lending institutions is high. Cisko and Kliestik define credit risk as the chances of delay or failure in payment or repay a loan by bank customers [3]. Credit risk, especially credit card default is a big challenge for banking/lending institutions. It poses a threat to the banking system as the losses incurred might be hard to revert. Even though there are tough measures against defaulting, it is still prevalent across the globe and the percentage of defaulters in increasing rapidly. This calls the need to have accurate default prediction models. Gauging the nature of a customer, whether the customer will adhere to all the banking rules, is a complex problem. Credit card assessment system is vital in reasonably predicting whether the borrowers will be able to repay their loan [4]. Banks generally categorize customers into two groups: good customers that have a high probability to pay monthly bills and repay loans on time, and bad customers who do not do so [5].

Recently, data mining, as part of Business Intelligence (BI) systems, is picking up a lot of traction with advances in research in both the domains. It assesses the predictive power of different algorithms/models used and provides insights that would be practically impossible to get if traditional, time-consuming, statistical methods are used. This technique is particularly useful in banking and financial sectors as it enables to solve problems that are impossible to predict manually due to availability of large body of information. The objective of this study is to build a model based on ML techniques that can predict if a customer can be a potential defaulter. We have carried out extensive analysis that compares different models based on their performances.

The paper is organized as follows: section II contains review of literature, section III discusses methodology, sections IV discusses exploratory data analysis techniques, section V contains model implementation, results are discussed in section VI and conclusions of the work in section VII.

II. Review of Literature

Abbas Keramati et.al [6] have carried out exhaustive literature survey pertaining to application of machine learning techniques in credit risk evaluation context. Their aim is to review each method comprehensively and find limitations of each method in order to come up with a model with better capabilities. They have reviewed 10 data mining techniques comprehensively and have proposed a hybrid model with better capabilities. They observed that hybrid SVM models will yield better results. Vasilica Oprea et.al [7] have assessed different techniques of data extraction and analysis for their accuracy and have also calculated the probability of default. TUDOR et.al [8] have tried to solve the same challenge by using clustering techniques. Customers were grouped using kNN clustering technique based on attribute similarity. However, authors haven't carried out extensive clustering accuracy analysis.

978-1-7281-8416-6/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: Carleton University. Downloaded on May 31,2021 at 12:48:26 UTC from IEEE Xplore. Restrictions apply.

III. Proposed Methodology

The methodology we have proposed to detect online credit card fraudulent transactions is illustrated in Figure 1:

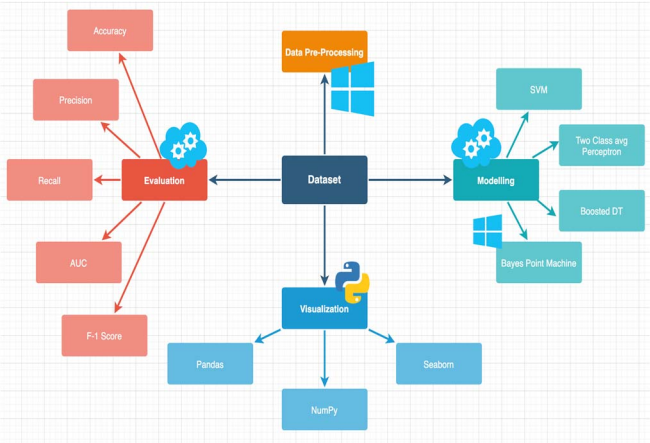

Fig. 1. Proposed Methodology for predicting Defaulters

A. Dataset Description

We have obtained this dataset [22] from UCI ML Repository. The default of credit card clients dataset has 30000 observations and 24 features. The records are from a Taiwanese bank/credit card issuer. and This dataset has been widely used by many researchers. The feature set of the data has demographic information (like: age, gender, marriage, education level) prior bill amount and prior payment history. In this dataset nearly 22% cardholders have a history of default payment.

B. Cleaning the Dataset

We have made the dataset appropriate to be used in training phase. This step involved renaming appropriate column names, grouping unknows in features: 'education' and 'marriage' and assigning to others.

C. Data Analysis

Our methodology focusses on trying to understand the default of credit card clients dataset and come up with an effective model to predict credit card defaulters. To get more insights on the dataset we have performed exploratory data analysis (EDA) using widely used open source libraries such as NumPy, Pandas, matplotlib, Seaborn, etc.

D. Azure Machine Learning (Azure ML) Platform

Microsoft Azure ML is a cloud-based platform for creating and managing end-to-end ML solutions. It's designed to help ML engineers and data scientists to leverage their existing skills and model frameworks. It also enables them to scale, distribute and deploy their workloads to the cloud [21]. Users can perform a variety of tasks like importing the dataset, training, splitting, clustering and multitude of other exploratory and ML tasks from the convenience of a simple web browser. Azure ML has built-in support for some of widely used open-source python frameworks. It has all the tools to create an end-to-end ML pipeline.

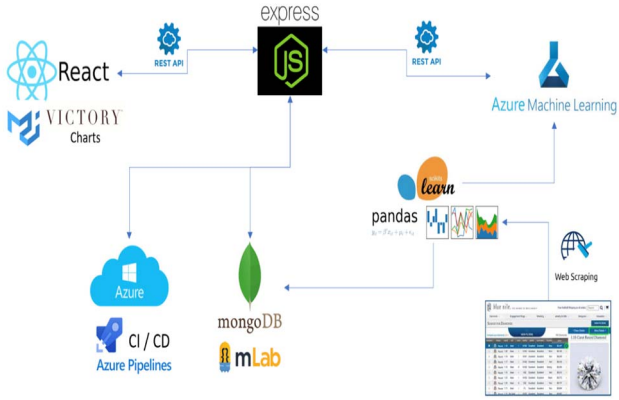

Fig. 2. Azure Machine Learning [20]

E. ML Algorithms

- Two Class Averaged Perceptron:

Perceptron classifies the given input data. It is a linear binary classifier as it classifies input data into two parts. Perceptrons are the foundations of neural networks. It is also used in supervised learning [11]. A linear function classifies the inputs into several possible outputs in the averaged perceptron method. A linear function classifies the inputs to several possible outputs. These classified outputs are then combined with a set of weights which are derived from the feature vector. Deep neural networks can model more complex boundaries, for linearly separable case a simpler perceptron model will do the job. For continuous training we can use perceptron as they are very fast and process input data serially [13].

- Two Class BDT:

DT has wide area of influence in machine learning, covering both classification and regression. A decision tree as the name goes uses a tree-like model and can be used to visually represent decisions. DTs are widely used in machine learning [9]. As the Boosted DTs are ensemble learning methods in which the upcoming successive trees rectify the errors of all the prior trees. The entire ensemble of trees together determines the final output. Generally Boosted DTs, give very good performance on a range of ML tasks. However, the downside is they are memory-intensive learners. Thus, a boosted DT model will not be able to handle very large datasets as the current implementation is held in memory [10].

- Two Class Deep Support Vector Machine:

SVMs try to find an optimal hyperplane that very clearly classifies data points in an N-dimensional feature space. There can be infinite number of hyperplanes that separate two classes. SVMs try to find a hyperplane that gives a maximum margin of separation (maximum distance between data points belonging to different classes) [13]. SVMs are amongst the earliest of machine learning algorithms, and SVM models find applications in wide ranging fields. We can use SVMs for both classification and regression tasks. In the training process, the algorithm recognizes patterns in N-dimensional feature space by analyzing the input data. All instances are mapped to this N-dimensional space and are subsequently mapped to output categories in such a way that there is maximum separation/gap

Authorized licensed use limited to: Carleton University. Downloaded on May 31,2021 at 12:48:26 UTC from IEEE Xplore. Restrictions apply.

between the categories. New sample points are assigned to one category or other during the prediction phase [14].

- Two Class Bayes Point Machine:

Using SVMs, BPM can be converted to a nonlinear classifier. It achieves good generalization performance by approximating the Bayesian average [16, 17]. This algorithmic implementation follows a Bayesian approach to linear classification called BPM. It generally doesn't overfit the training data [18].

IV. Exploratory Data Analysis (EDA)

We have carried out EDA to get some insight on the dataset.

Fig. 3. Class Distributions

In the above figure, "0" represents clients who will not default and "1" represents clients who are going to default. From the above graph, we can see that there are 23364 non-defaulters and 6636 defaulters.

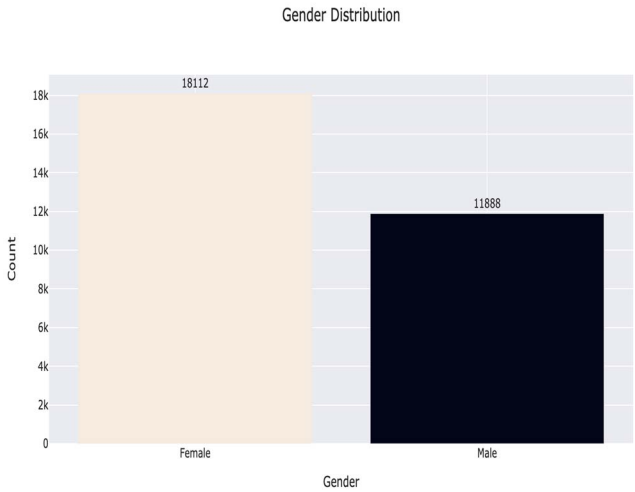

Fig. 4. Gender Distribution Plot

From histogram plot of Figure 4, we can infer that the bank has more female customers than male customers.

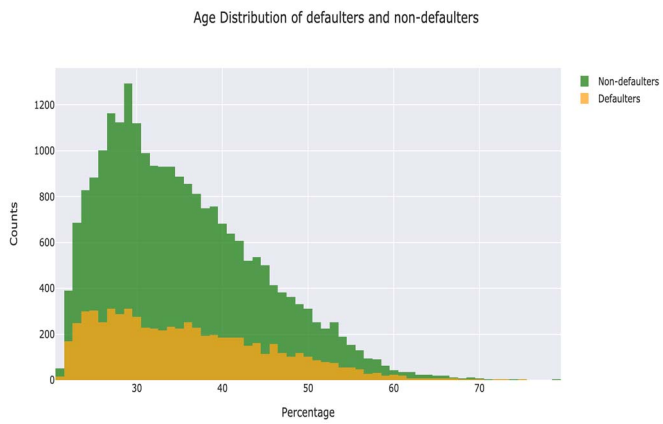

Fig. 5. Age Distribution of Defaulters vs non-Defaulters

From the above overlaid histogram plot we can say that greater is the age of the customer lesser are the chances for that customer to be a defaulter. Also, we can infer that this credit card company has maximum customers in the age range of 25-35 years.

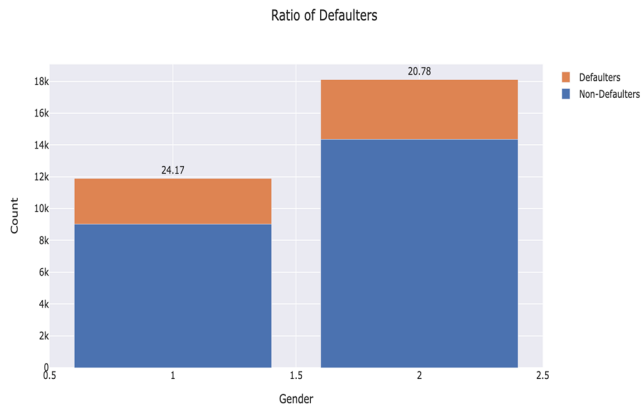

Fig. 6. Ratio of Defaulters for 'Gender' Feature

Correlation gives us an indication of the extent to which two or more features fluctuate together. In other words, it gives us an indication as to how much does a change in one variable/feature causes for a slight change in another variable/feature. It may be positive, negative or zero depending on the nature of relationships.

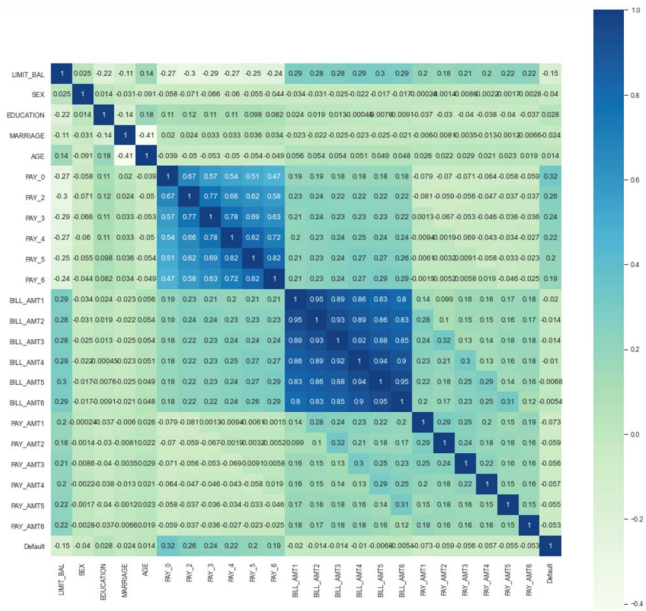

Fig. 7. Correlation Matrix

Authorized licensed use limited to: Carleton University. Downloaded on May 31,2021 at 12:48:26 UTC from IEEE Xplore. Restrictions apply.

From the above correlation matrix we can infer that there is a positive correlation between the 'payment' and 'bill amount' features.

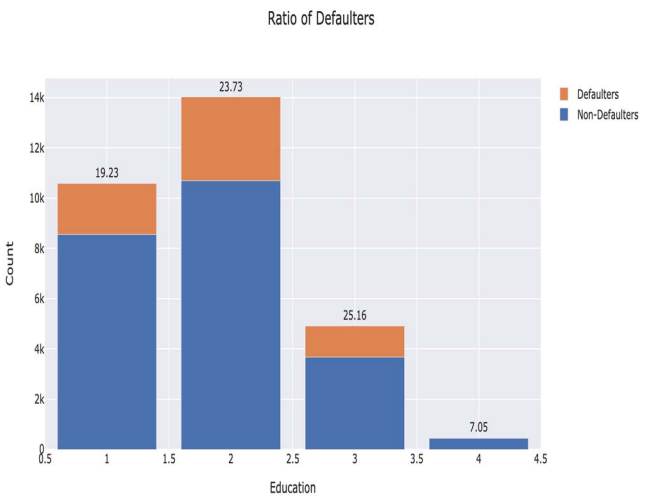

Fig. 8. Defaulters ratio for feature 'Education'

V. Model Implementation

SVM, BDT, BPM and AP algorithms were used to train a model that can detect credit card defaulters. We have trained our models on Azure ML platform. We have evaluated the performance of our trained models using the following metrices: AUC, Accuracy, F-1 Score, Precision and Recall. We have computed the Confusion matrix for all the models.

Parameters used for computing Confusion Matrix were:

- True Positive (TP): correctly predicts a defaulter as a defaulter.
- True Negative (TN): correctly predicts a non-defaulter as non-defaulter.
- False Positive (FP): predicts a non-defaulter as defaulter.
- False Negative (FN): predicts a defaulter as non-defaulter.

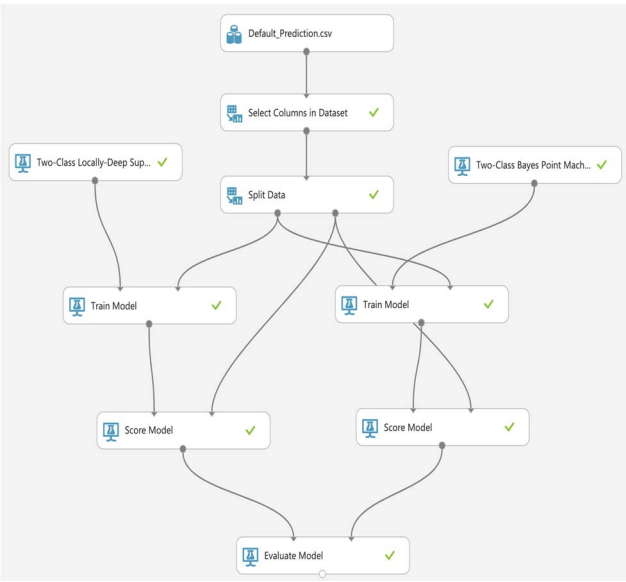

Fig. 9. Azure Machine Learning model

Following are the metrics of evaluation:

- Precision = TP / (TP+FP)
- Recall (Sensitivity) = TP / (TP+FN)
- Accuracy = (TP+TN) / (TP+FP+FN+TN)
- F1 score = 2*(Recall * Precision) / (Recall + Precision)
- Area under ROC Curve (AUC):

  • AUC value of 1 is a perfect score.
  • Higher value of AUC indicates better performance of the classifier.

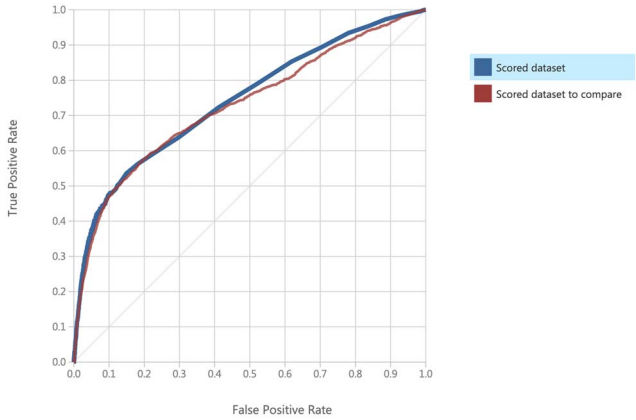

Fig. 10. ROC Curve for SVM and BPM

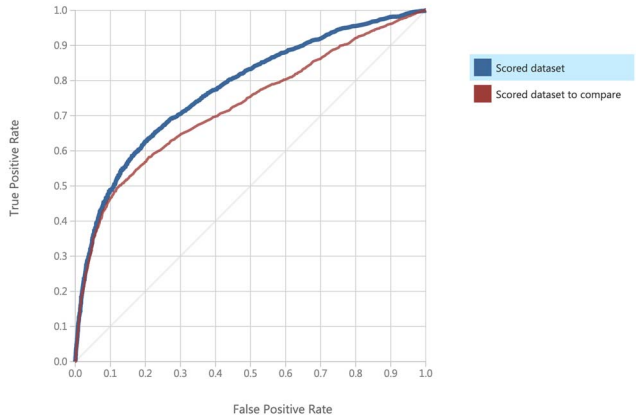

Fig. 11. ROC Curve for BDT and AP

VI. Results

We have accurately predicted credit card defaulters by training ML models using BDT, BPM, DSVM and AP algorithms on Azure Machine Learning platform. The accuracy and other metrics of evaluation are evident from the above tables. The performance of model using DSVM is better than all other models. It is giving us an overall accuracy of 82.2% and AUC of 0.74.

Authorized licensed use limited to: Carleton University. Downloaded on May 31,2021 at 12:48:26 UTC from IEEE Xplore. Restrictions apply.

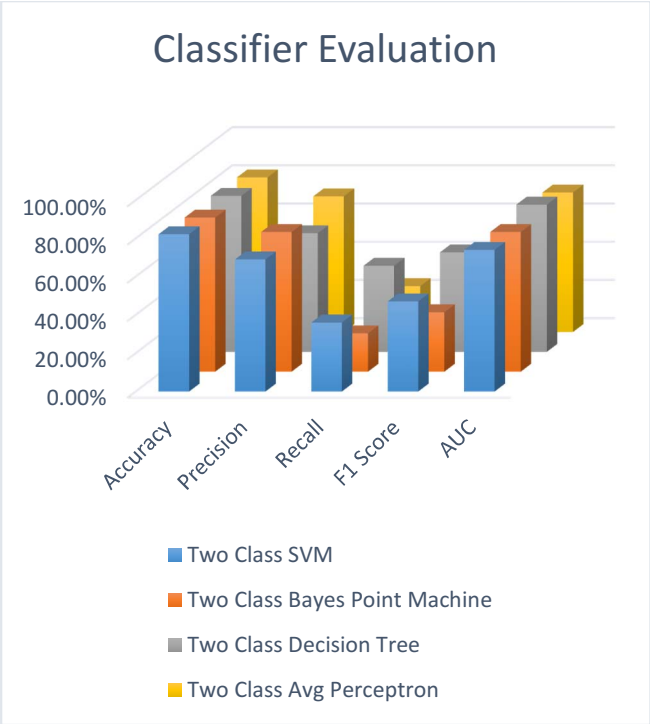

Fig. 9. Classifier Evaluation

TABLE I. CONFUSION MATRIX

	Accuracy	Precision	Recall	F1 Score	AUC
Two Class SVM	82.20%	0.69	0.36	0.47	0.74
Two Class Bayes Point Machine	80.60%	0.73	0.2	0.31	0.73
Two Class Decision Tree	81.60%	0.62	0.45	0.52	0.77
Two Class Avg Perceptron	80.90%	0.71	0.24	0.36	0.73

TABLE II. CLASSIFICATION RESULTS

	TP	TN	FP	FN
Two Class SVM	717	6679	330	1274
Two Class Bayes Point Machine	394	6863	146	1597
Two Class Decision Tree	887	6455	554	1104
Two Class Avg Perceptron	467	6814	195	1524

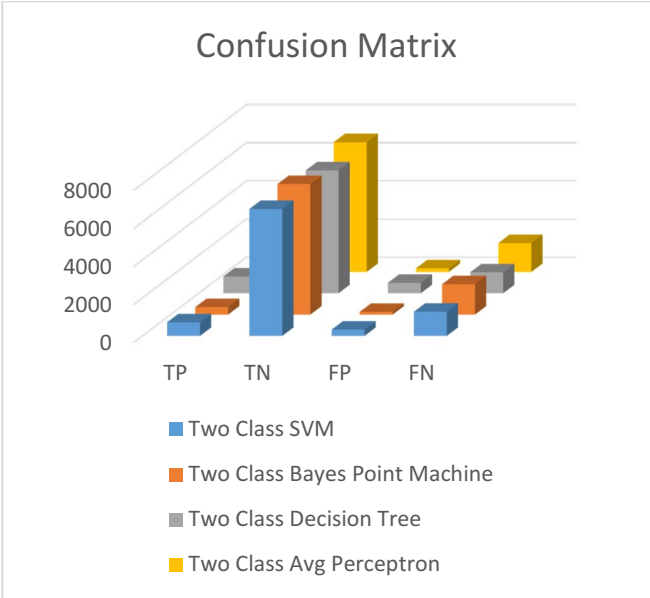

Fig. 10. Confusion Matrix

VII. Conclusion and Future Work

It is crucial for banks/lending institutions to predict defaulters. In this work, we have designed a defaulter detection technique which identifies credit card defaulters accurately. DSVM has shown promising results to be adapted in defaulter detection systems. Future work will involve training defaulter detection system using Multilayer Perceptron.

References

[1] Tan, Pang-Ning, Steinbach, Michael, Kumar, Vipin, 2005. Introduction to Data Mining, first ed. Addison-Wesley Longman Publishing Co. Inc., Boston, MA, USA

[2] "Data Mining & Machine Learning", Jun 2019, [online] Available: https://bernardmarr.com/default.asp?contentID=1741

[3] Cisko Š, Klieštik T. Finančný manažment podniku II. Žilina (2013) EDIS Publishers, University of Žilina.

[4] Mandala I.G.N.N., Nawangpalupi C. A., and Praktikto F. R (2012) Assessing Credit Risk: An Application of Data Mining in a Rural Bank. Procedia Economics and Finance. 4:406–412.

[5] Akkoc S (2012) An empirical comparison of conventional techniques, neural networks and the three stage hybrid adaptive neuro fuzzy inference system (ANFIS) model for credit scoring analysis: the case of Turkish credit card data. Eur J Operat Res 222(1):168–178.

[6] Abbas Kerama and Niloofar Yousefi - A Proposed Classification of Data Mining Techniques in Credit Scoring. Proceedings of the 2011,International conference on industrial engineering and Operations management Kuala Lumpur, Malaysia.

[7] Tudor, Adela Ioana, Adela Bâra, and Simona Vasilica Oprea. "Comparative analysis of data mining methods for predicting credit default probabilities in a retail bank portfolio." WSEAS International Conference. Proceedings. Recent Advances in Computer Engineering Series. No. 7. WSEAS, 2012.

[8] Tudor, Adela Ioana, Adela Bara, and Elena Andrei. "Clustering analysis for credit default probabilities in a retail bank portfolio." Bucharest Acad Econ Stud, Database Syst J 3 (2012): 23-30.

[9] "Decision Trees in Machine Learning", Nov 2018, [Online] https://towardsdatascience.com/decision-trees-in-machine-learning-641b9c4e8052

[10] "Two-Class Boosted Decision Tree" Jan 2019, [Online] https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/two-class-boosted-decision-tree

[11] "What is perceptron?", Sep 2019[Online] https://towardsdatascience.com/what-is-a-perceptron-210a50190c3b

Authorized licensed use limited to: Carleton University. Downloaded on May 31,2021 at 12:48:26 UTC from IEEE Xplore. Restrictions apply.

[12] "Azure Machine Learning", Jan 2020, [Online] https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/two-class-averaged-perceptron

[13] "Two Class Average Perceptron", May 2019, [online] Available: https://medium.com/datadriveninvestor/6-reasons-microsoft-has-become-the-go-to-for-machine-learning-e642864ef5f5

[14] "Support Vector Machine — Introduction to Machine Learning Algorithms", Jun 2018, [online] Available: https://towardsdatascience.com/support-vector-machine-introduction-to-machine-learning-algorithms-934a444fca47

[15] "Two-Class Support Vector Machine", May 2019, [online] Available: https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/two-class-support-vector-machine

[16] R. Herbrich, T. Graepel, and C. Campbell. Bayes point machine: Estimating the Bayes point in kernel space. In IJCAI Workshop SVMs, pages 23–27, 1999.

[17] T. Minka. A family of algorithms for approximate Bayesian inference. PhD thesis, MIT, Jan. 2001. www.media.mit.edu/~tpminka/papers/learning.html.

[18] "Two-Class Bayes Point Machine", Mar 2020, [online] Available: https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/two-class-bayes-point-machine

[19] "Understanding Feature extraction using Correlation Matrix and Scatter Plots", [online] Available: https://towardsdatascience.com/understanding-feature-extraction-using-correlation-matrix-and-scatter-plots-6c19e968a60c

[20] "Full Stack Machine Learning on Azure", [online] Available: https://towardsdatascience.com/full-stack-machine-learning-on-azure-f0f6b77be07e

[21] "Azure Machine Learning Service", [online] Available: https://towardsdatascience.com/azure-machine-learning-service-part-1-an-introduction-739620d1127b

[22] "Default of Credit Card Clients Data Set", [online] Available: https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients

Authorized licensed use limited to: Carleton University. Downloaded on May 31,2021 at 12:48:26 UTC from IEEE Xplore. Restrictions apply.